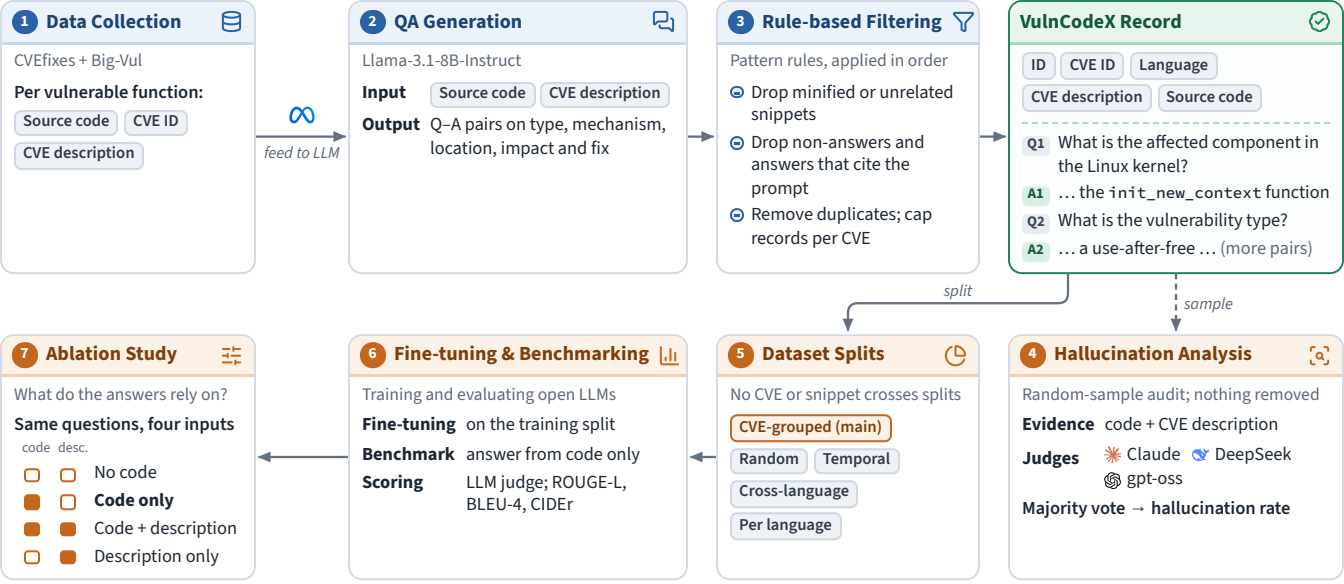

1 Data Collection
CVEfixes + Big-Vul
Per vulnerable function:
Source code
CVE ID
CVE description
feed to LLM
2 QA Generation
Llama-3.1-8B-Instruct
Input
Source code
CVE description
Output
Q–A pairs on type, mechanism, location, impact and fix
3 Rule-based Filtering
Pattern rules, applied in order
Drop minified or unrelated snippets
Drop non-answers and answers that cite the prompt
Remove duplicates; cap records per CVE
VulnCodeX Record
ID
CVE ID
Language
CVE description
Source code
Q1
What is the affected component in the Linux kernel?
A1
… the init_new_context function
Q2
What is the vulnerability type?
A2
… a use-after-free … (more pairs)
split
sample
7 Ablation Study
What do the answers rely on?
Same questions, four inputs
code
desc.
No code
Code only
Code + description
Description only
6 Fine-tuning & Benchmarking
Training and evaluating open LLMs
Fine-tuning
on the training split
Benchmark
answer from code only
Scoring
LLM judge; ROUGE-L, BLEU-4, CIDEr
5 Dataset Splits
No CVE or snippet crosses splits
CVE-grouped (main)
Random
Temporal
Cross-language
Per language
4 Hallucination Analysis
Random-sample audit; nothing removed
Evidence
code + CVE description
Judges
Claude
DeepSeek
gpt-oss
Majority vote → hallucination rate